

The Pendulum

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Abstract:

In this experiment, we modelled the motion of a simple pendulum, a damped oscillator, and a damped driven oscillator, using linear and non-linear approximations for the equations of motion. The data obtained was then plotted in gnuplot. It was concluded that the linear approximation is always sufficiently accurate enough for small initial angles, but for larger initial angles the non-linear approximations were needed. It was also concluded that while using the trapezoid rule to obtain the equations of motion was acceptable for the simple pendulum, that this was not the case for the damped pendulum, for which the Runge-Kutta method was far more accurate. Finally, the period doubling and chaotic behaviour that occurs as the amplitude of the driving force of a damped driven pendulum is increased, was observed through phase portraits.

Aims:

The aims of this experiment were;

- To investigate the equation of motion of a simple pendulum
- To investigate the equation of motion of a damped pendulum
- To investigate the behaviour of a damped driven pendulum
- To investigate the numerical solutions of differential equations using the trapezoid rule and the Runge-Kutta method
- To learn how to write output data to a file in C programming
- To learn how to use gnuplot to create graphs from datafiles

Introduction and Background Theory:

The Simple Pendulum

According to Newton's Second Law of motion, the equation for the pendulum is

$$\frac{d^2 s}{dt^2} = L \frac{d^2 \theta}{dt^2} = -g \sin \theta$$

Where L is the length of the pendulum, and $g \sin \theta$ is the component of the acceleration due to gravity in the downward direction.

This is a non-linear equation, but may be linearised by using the approximation that, for small angles, $\sin \theta \approx \theta$. We then obtain the following,

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{L} \theta$$

This equation can then be solved for the usual solution for a harmonic oscillator,

$$\theta = A \sin(\beta t + \phi), \quad \beta = \sqrt{\frac{g}{L}}$$

Where A is the amplitude, and ϕ is the constant (initial) phase.

Second Order Differential Equations

To solve a second order differential equation numerically, one must introduce a new variable and transform the second order equation into two first order differential equations. This may be performed in both the linear and non-linear cases, by using the angular velocity of the bob, ω , which is defined as

$$\omega = \frac{d\theta}{dt}, \quad \frac{d\omega}{dt} = f(\theta, \omega, t) = -\frac{g}{L} \sin \theta$$

The Simple Euler Method

The Euler methods for solving the simple pendulum differential equations involves choosing initial values for θ and ω , then using a Taylor expansion to find the values of these after some time has passed. Disregarding terms of second order or higher, we are left with the following expansions;

$$\theta_{n+1} = \theta_n + \omega_n \Delta t$$

$$\omega_{n+1} = \omega_n - \frac{g}{L} \theta_n \Delta t$$

There are error terms of order $(\Delta t)^2$ and higher.

The Trapezoid Rule

The trapezoid rule makes improved approximations for updating the angular frequency and angle, by approximating the area under the curve as a trapezoid. This is accurate the second order for a timestep h , but this is not sufficiently accurate for a damped driven pendulum.

The Runge-Kutta Method

The fourth order Runge-Kutta algorithm is a linear approximation given by

$$\theta(t+h) = \theta(t) + (k_{1a} + 2k_{2a} + 2k_{3a} + k_{4a})/6$$

$$\omega(t+h) = \omega(t) + (k_{1b} + 2k_{2b} + 2k_{3b} + k_{4b})/6$$

In these equations, the k's are the values for the slope of the function at points across the interval. This algorithm is accurate to the 4th order terms, hence it is called the fourth order Runge-Kutta.

The Damped Driven Oscillator

The model above is only accurate for a simple pendulum, however this can be extended to account for damping and driving forces, given by $-mk\omega$ and $mA \cos(\Omega t)$ respectively. This transforms our equation of motion to the following;

$$\frac{d\omega}{dt} = f(\theta, \omega, t) = -\beta^2 \sin \theta - k\omega + A \cos(\Omega t)$$

Where k is a damping coefficient, A is the amplitude of the driving force, and Ω is the angular frequency of this force.

This formula can once again be linearised using the small angle approximation to obtain

$$\frac{d\omega}{dt} = f(\theta, \omega, t) = -\beta^2 \theta - k\omega + A \cos(\Omega t)$$

Experimental Method:

A C program was used to simulate the system of the pendulum, and to write the data to a file. The data was then graphed. This was performed for a number of cases;

- i. The simple pendulum, for both the linear and non-linear equations of motion using the trapezoid rule
- ii. The simple pendulum, for both the linear and non-linear equations of motion using the 4th order Runge-Kutta method
- iii. The damped oscillator for both the linear and non-linear equations of motion using the 4th order Runge-Kutta method
- iv. The damped driven oscillator for both the linear and non-linear equations of motion using the 4th order Runge-Kutta method

In each case, the starting conditions that were used were $\theta = 0, 0.2, 1, 3.124$ and $\omega = 1, 0, 0, 0$. A variety of graphs were then plotted using gnuplot for each case.

Results:

Case i) The Simple Pendulum using the Trapezoid Rule

Linear Approximation:

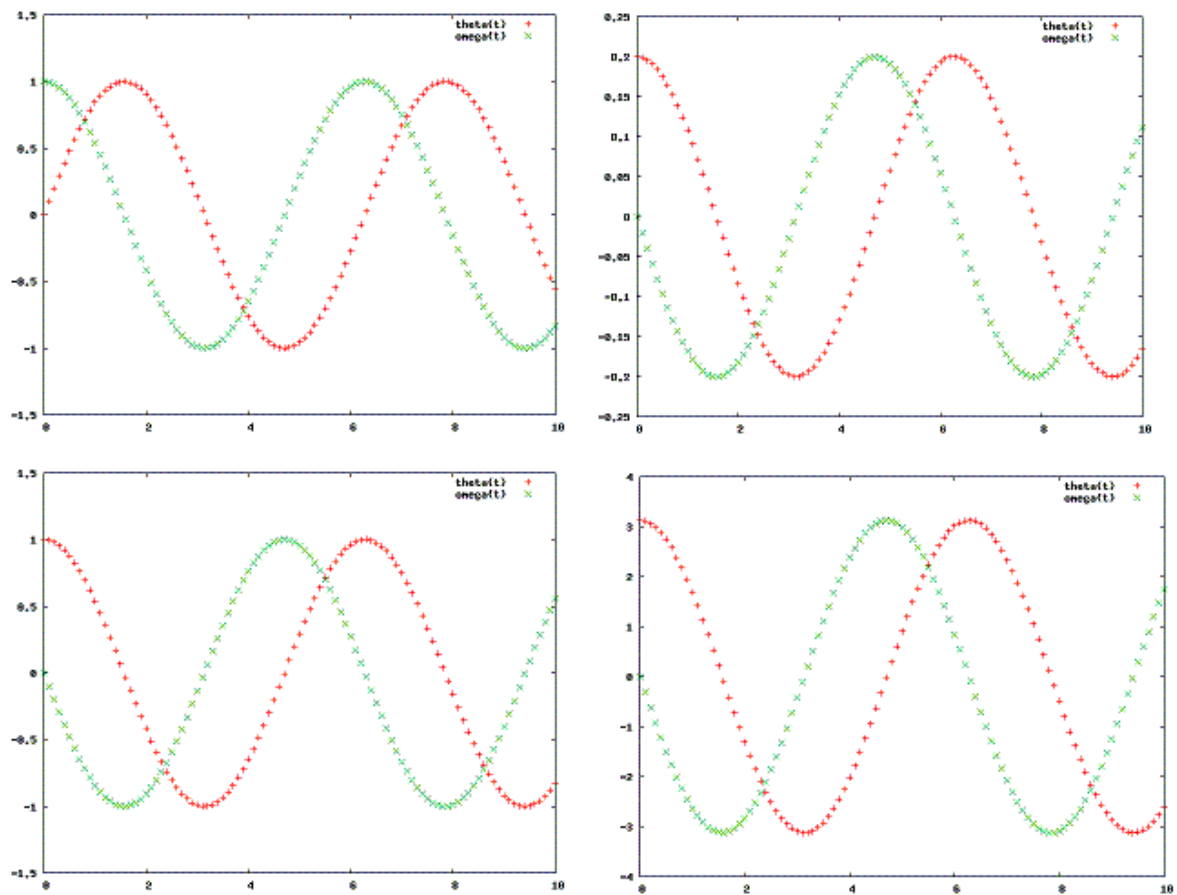


Fig 1. Graphs of $\theta(t)$ and $\omega(t)$ for various starting values

The linear approximation gives adequate values for small initial angles, but for large starting angles the results are still a perfect sinusoid, which is inaccurate.

Non-linear Approximation:

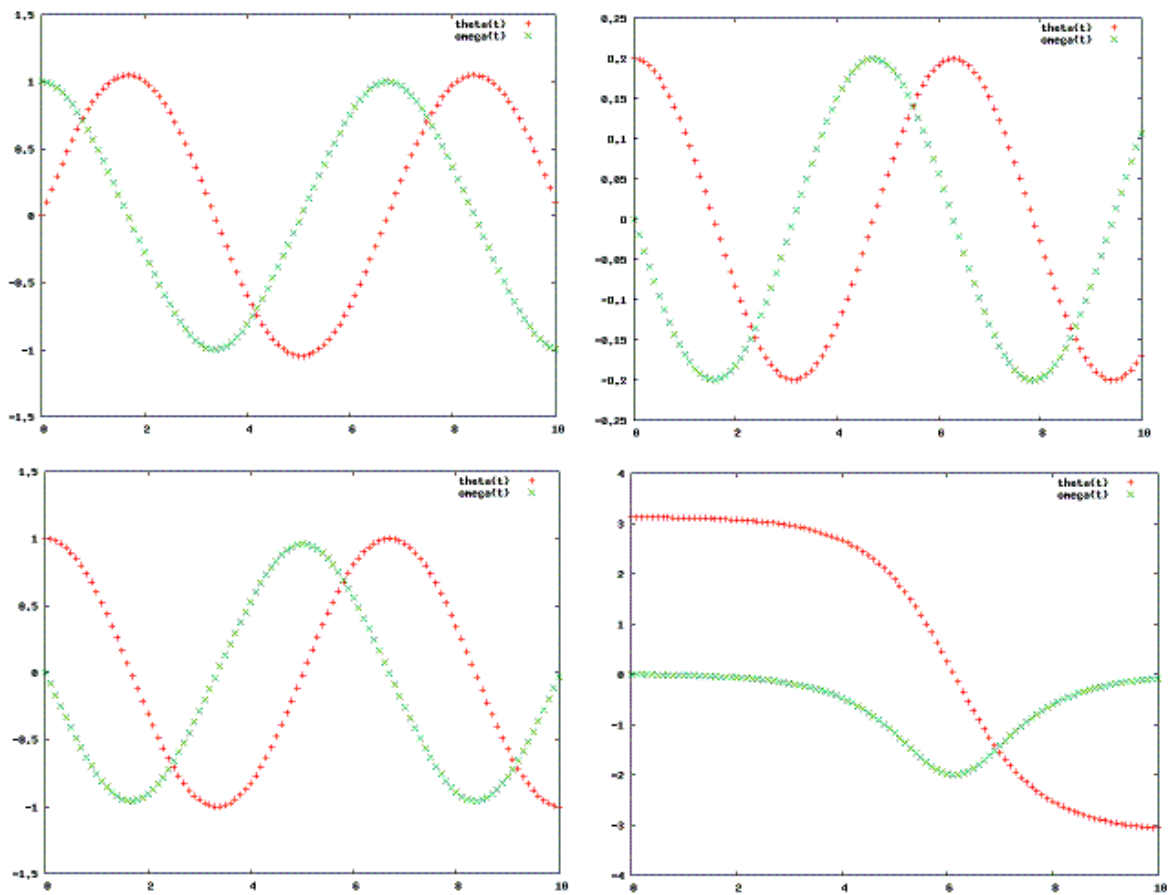


Fig. 2 Graphs of $\theta(t)$ and $\omega(t)$ for various starting values

For small starting angles and large starting angles very different behaviour is observed. The non-linear approximation is far more accurate for large initial angles.

Comparison of Linear and Non-linear Approximations:

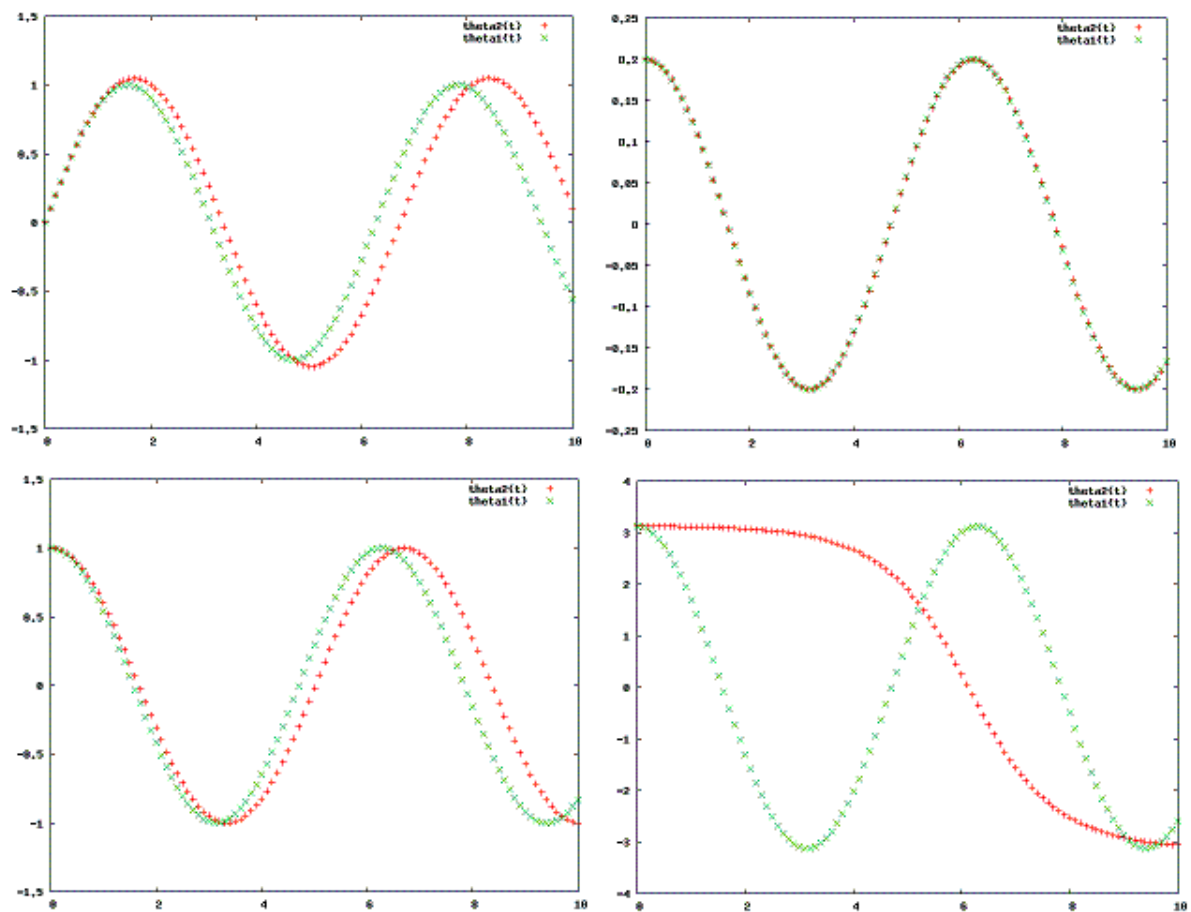


Fig. 3 Comparison of linear $\theta(t)$ [in red] and non-linear $\theta(t)$ [in green]

In these diagrams it can be seen that as the initial angle increases, the linear approximation becomes increasingly incorrect. The linear approximation only yields a sinusoidal picture because of the small angle approximation, whereas the non-linear approximation does not. This can be seen, for example, in the last case, which is modelling the behaviour when a simple pendulum is released from almost vertical.

Case ii) The Simple Pendulum using the 4th Order Runge-Kutta Method

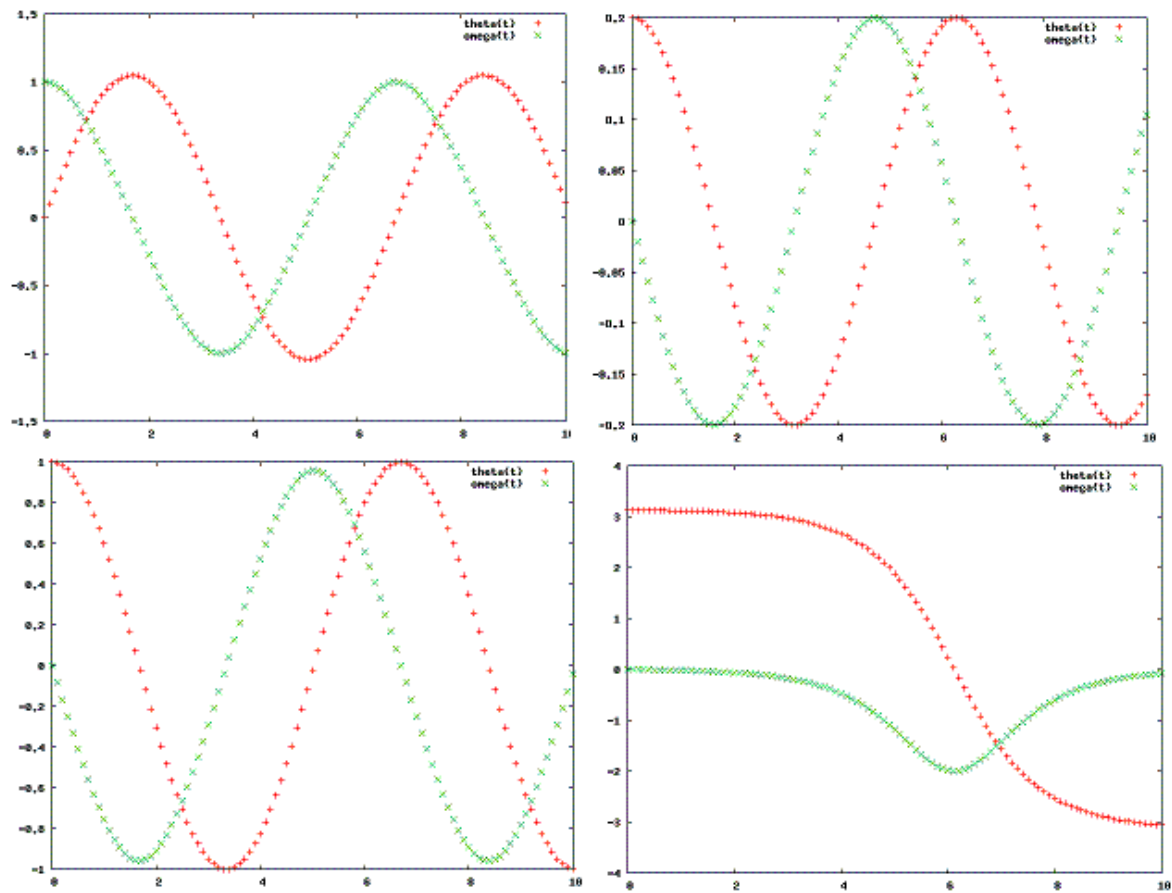


Fig. 4 Graphs of $\theta(t)$ and $\omega(t)$ for various starting values

When the data obtained from the Runge-Kutta non-linear approximation was plotted, and compared to the non-linear approximation using the trapezoid rule it was observed that there was no difference in accuracy for a simple pendulum, as is shown by the similarity between the figures above and those for the non-linear approximation in case i.

Case iii) The Damped Oscillator using the 4th Order Runge-Kutta Method

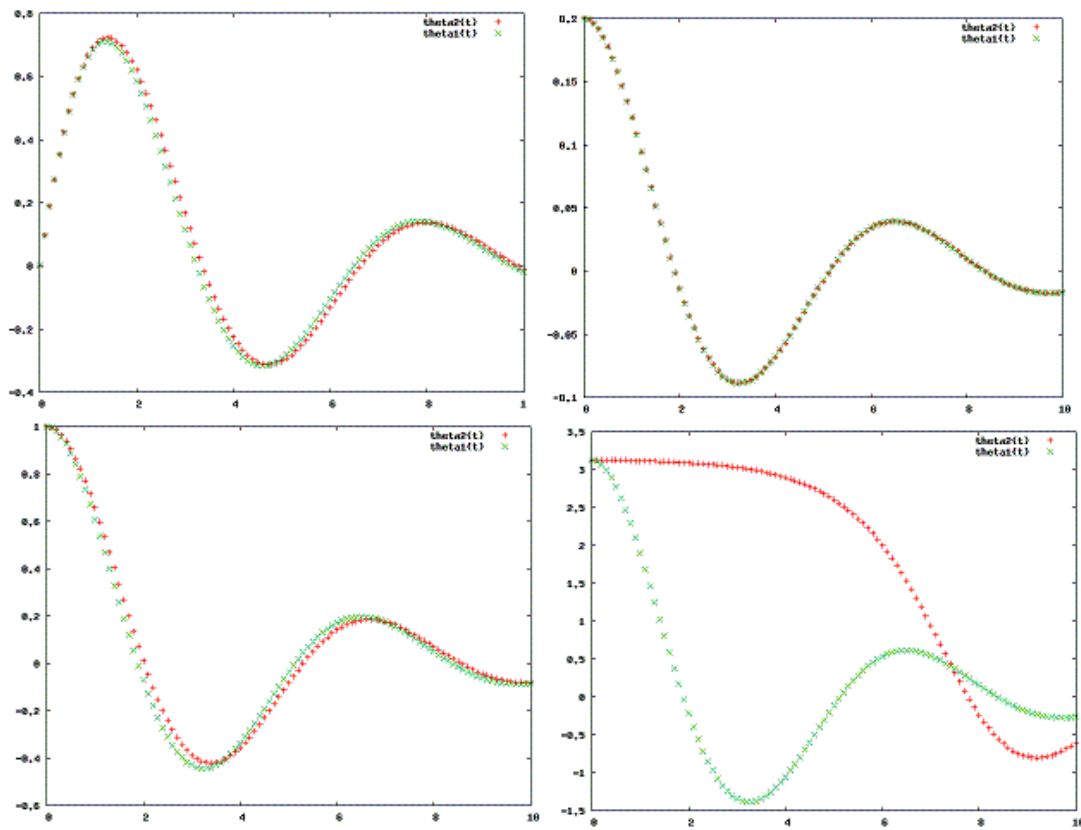


Fig. 5 Comparison of linear $\theta(t)$ [in red] and non-linear $\theta(t)$ [in green] with damping

From the comparison of linear and non-linear approximations for the damped oscillator, we can see once again that for small starting angle, the linear approximation is sufficiently accurate, but for larger initial angles this becomes less accurate. However, both approximations will eventually converge to the same final value near zero as static equilibrium is reached, as shown below.

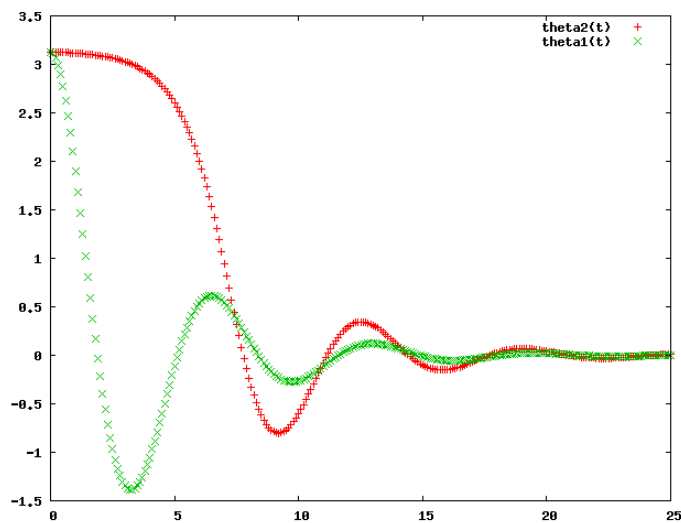


Fig. 6 Comparison of linear and non-linear behaviour over a longer timescale

Case iv) The Damped Driven Oscillator using the 4th Order Runge-Kutta Method

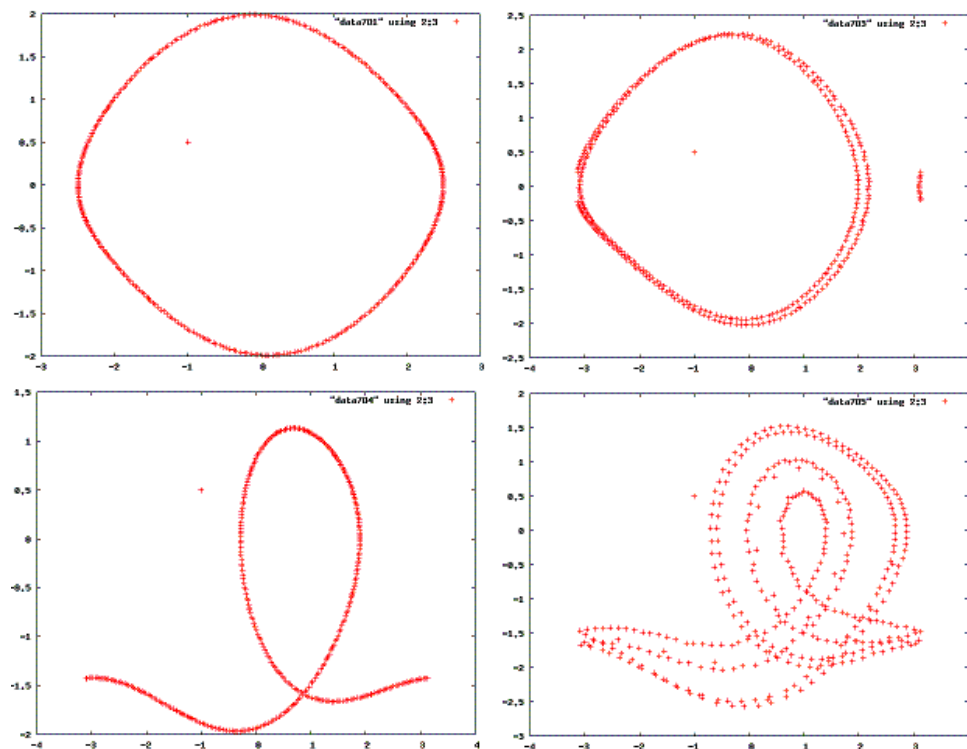


Fig. 7 Phase portraits showing the behaviour of a damped driven oscillator

The figures above are phase portraits describing the behaviour of a damped driven oscillator, with θ plotted against ω , for the following amplitudes, $A = 0.9, 1.07, 1.35, 1.47$. It can be seen from the above diagrams that when the amplitude is sufficiently large, that period doubling occurs. When the amplitude reaches a value of 1.5, chaotic behaviour can be observed over a longer time interval, as seen below.

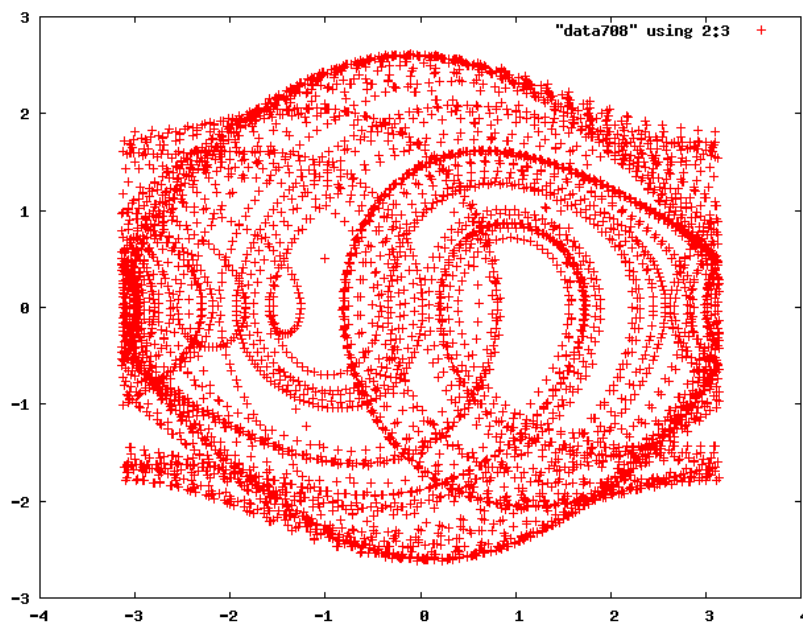


Fig. 8 Chaotic behaviour observed when $A = 1.5$

Discussions and Conclusions:

It can be concluded from the simulations above that for both the simple pendulum and the damped pendulum, and for small initial angles, the linear approximation is adequate. However, as the angle is increased, models such as the non-linear approximation are needed to maintain accuracy, although for the damped oscillator both the linear and the non-linear approximations converged to the same value eventually.

It was shown that for small angles, the trapezoid rule and the Runge-Kutta method were equally accurate, but for larger angles the Runge-Kutta method was a far superior algorithm. Using the Runge-Kutta method, it was possible to create phase portraits to show the period doubling and chaotic behaviour that occurs as the amplitude of the driving force of a damped driven pendulum is increased.